Optimizing Communication Performance for Smart IoT by Integrating UAV and Reconfigurable Intelligent Surface: A Deep Reinforcement Learning Approach

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***Abstract*—With the proliferation of smart Internet of Things (IoT) applications, providing reliable wireless connectivity for numerous users in structurally complex urban environments has become a core challenge for next-generation communication networks. Traditional terrestrial base stations are susceptible to building blockages, leading to non-line-of-sight (NLOS) transmission issues that severely degrade signal quality. To address this, this study proposes a cooperative control system that integrates an Unmanned Aerial Vehicle (UAV), a Reconfigurable Intelligent Surface (RIS), and Deep Reinforcement Learning (DRL) to dynamically optimize data collection and communication performance in urban settings. The key contribution of this research lies in the design and implementation of an intelligent network deployment framework that more closely mirrors reality. First, we introduce a user clustering mechanism based on the DBSCAN algorithm to automatically group ground-based IoT devices. Subsequently, Voronoi diagrams are used to generate candidate anchor points for the RIS, ensuring a rational deployment. Building on this framework, we model the problem of a UAV-mounted RIS serving specific user clusters at predefined anchor points as a DRL optimization task aimed at maximizing the system's sum-rate. We systematically evaluate several advanced DRL algorithms, including Soft Actor-Critic (SAC), Deep Deterministic Policy Gradient (DDPG), Proximal Policy Optimization (PPO), and a novel Beta-Space Exploration SAC designed for hardware non-idealities. To ensure effective training, we devised a "virtual user complementation mechanism," enabling the model to learn stably within a fixed action and state space, regardless of the number of real users being served. Experimental results demonstrate that in our challenging NLOS-dominant scenarios, all DRL-based methods significantly outperform a random phase baseline. Notably, the Beta-Space Exploration SAC algorithm exhibits the best performance and robustness, achieving an average sum-rate of 3.35 bps/Hz with an L=64 RIS configuration, markedly surpassing standard SAC, DDPG, and PPO. This study not only validates the feasibility of the UAV-RIS system but also provides a complete framework—from environment generation, model training, to performance evaluation—offering crucial technical reference and empirical evidence for the adaptive deployment of future smart city communication networks.**

***Index Terms*—Reconfigurable Intelligent Surface (RIS), Deep Reinforcement Learning (DRL), Unmanned Aerial Vehicle (UAV), User Clustering, DBSCAN, NLOS Communication, Resource Optimization.**

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I. Introduction

With the rapid development of Internet of Things (IoT) technologies and the progressive realization of smart city concepts, providing efficient and reliable wireless connectivity for a massive number of dispersed IoT devices in structurally complex urban environments has become a core challenge for current and future communication networks [1], [2]. Conventional terrestrial Base Station (BS) deployments in densely built-up areas often suffer from severe signal blockage, leading to Non-Line-of-Sight (NLOS) propagation predicaments that degrade communication quality and create coverage holes [3].

To overcome the physical limitations of ground-based infrastructure, Unmanned Aerial Vehicles (UAVs) are considered a highly promising solution for aerial communication platforms due to their excellent mobility, flexible deployment capabilities, and potential to establish Line-of-Sight (LoS) links [4], [5]. However, when relying solely on UAVs as aerial relays, the backhaul link between the UAV and the ground BS can still be obstructed by tall buildings, limiting overall system performance.

In recent years, the Reconfigurable Intelligent Surface (RIS) has emerged as a revolutionary 6G candidate technology, offering a new paradigm to address the aforementioned challenges [6], [7]. An RIS is composed of a large number of low-cost, passive reflecting elements that can be programmed via software to intelligently manipulate the phase and amplitude of incident electromagnetic waves. This capability allows for the reshaping of the wireless propagation environment to create high-quality virtual LoS links [8], [9]. By mounting an RIS on a UAV, a mobile and intelligent aerial reflecting platform is formed. This integration not only leverages the positional advantages of the UAV but also dynamically overcomes signal propagation obstacles in complex three-dimensional spaces, establishing reliable communication bridges for IoT devices in ground-level coverage holes [1].

However, the cooperative control of a UAV-RIS system presents a highly complex, non-convex optimization problem. Traditional mathematical optimization methods often struggle to find real-time, efficient solutions in dynamic environments [1], [6]. Deep Reinforcement Learning (DRL), with its powerful capabilities for environmental perception, autonomous decision-making, and policy learning, provides a potent tool

for tackling such problems [3]. Therefore, the core motivation of this research is to explore how advanced DRL algorithms can be utilized to solve the joint control problem of RIS phase-shifting and BS beamforming when a UAV is stationed at fixed anchor points. Our work not only considers realistic hardware limitations and channel uncertainties but also introduces a DBSCAN-based user clustering mechanism to simulate more authentic user distribution scenarios, with the overarching goal of maximizing the data collection efficiency of the entire IoT network.

The major contributions of this study are threefold. First, we propose an **innovative structured environment generation framework**. By employing the DBSCAN algorithm for user clustering and Voronoi diagrams for generating rational RIS anchor points, our framework creates structured, group-based service scenarios that mirror real-world deployments, thereby enhancing the DRL agent's training efficiency and the policy's practical applicability. Second, we provide a **comprehensive performance evaluation of advanced DRL algorithms**, including SAC, DDPG, PPO, and the novel Beta-Space Exploration SAC. This offers critical empirical insights into the performance hierarchy and suitability of these algorithms for the complex UAV-RIS optimization problem. Third, we develop and validate an effective **"Virtual User Complementation Mechanism"** to address the challenge of varying user group sizes, which ensures stable DRL model training by maintaining a fixed-dimensional state-action space and significantly improves the model's robustness and adaptability to practical deployment scenarios.

The remainder of this paper is organized as follows. Section II reviews the related literature and background on DRL algorithms. Section III details the system architecture and formulates the optimization problem. Section IV presents our proposed methodology, including the environment generation framework and DRL design. Section V presents and analyzes the experimental results. Finally, Section VI concludes the paper and discusses future research directions.

II. Related Work and Background

The synergistic integration of UAVs and RIS is a rapidly evolving field. This section situates our research at the forefront of current academic inquiry by systematically reviewing key advancements across several core themes: (1) advanced modeling and robustness, (2) dynamic trajectory and multi-agent coordination, (3) emerging application paradigms, and (4) integration with next-generation AI. Furthermore, we introduce the benchmark DRL algorithms used in this study.

A. Advances in UAV-RIS Cooperative Communications

1) Advanced Modeling and Robustness

Our work addresses the gap between idealized models and physical reality through our SAC-Mismatch and Beta-Space Exploration SAC algorithms. Recent literature further deepens this effort by tackling hardware impairments and channel uncertainty from multiple dimensions. Beyond the Phase-Dependent Amplitude (PDA) effect considered in our work, studies like Gurgunoglu et al. [10] have begun modeling more severe impairments such as phase noise, which degrades beamforming accuracy. Concurrently, robust optimization techniques are being employed to counteract the effects of imperfect Channel State Information (CSI), ensuring system performance even under worst-case error conditions [11]. To overcome the high overhead of traditional channel estimation, efficient methods leveraging the low-rank structure of RIS-assisted channels are being developed [12], aligning with our goal of creating physically-driven and efficient training environments.

2) Dynamic Trajectory and Multi-Agent Coordination

While our research focuses on resource optimization at fixed UAV anchor points, a significant line of research incorporates UAV trajectory planning as a key optimization variable. Jointly designing the UAV's trajectory and RIS configuration to maximize metrics like secrecy rate or reliability has been explored in [13]-[15]. To manage the high-dimensional action space of this joint problem, Hierarchical DRL (HDRL) has emerged as a powerful solution, decoupling the problem into long-term trajectory planning and short-term resource allocation [16]. When scaling to multi-UAV swarms, the problem shifts to multi-agent coordination, where Multi-Agent DRL (MADRL) provides the theoretical foundation [17], [18]. Advanced paradigms like Centralized Training with Decentralized Execution (CTDE) are critical for addressing the non-stationarity of multi-agent environments, balancing learning efficiency with real-time execution.

3) Emerging Application Paradigms

Our objective of sum-rate maximization is fundamental, but the diverse applications of smart cities demand a broader set of performance metrics. Key emerging paradigms include:

- **Energy Efficiency (EE) and SWIPT:** For battery-powered UAVs, maximizing the number of bits transmitted per Joule is critical. This is often coupled with Simultaneous Wireless Information and Power Transfer (SWIPT) to wirelessly charge low-power IoT devices [19], [20].
- **Age of Information (AoI):** For time-sensitive applications like traffic monitoring, minimizing the AoI to ensure data freshness is a primary objective, which often conflicts with maximizing throughput [21].
- **Mobile Edge Computing (MEC):** UAVs can serve as mobile MEC servers. Research in this area focuses on jointly optimizing task offloading decisions, UAV trajectory, and communication resources to minimize latency or energy consumption [17], [22].
- **Physical Layer Security (PLS):** To secure vulnerable air-to-ground links, PLS techniques aim to maximize the secrecy rate. More advanced covert communication seeks to hide the existence of the transmission itself [23], [24].

B. Background on DRL Algorithms

In addition to the SAC variants detailed in Section IV, we employ DDPG and PPO as important benchmarks.

1) Deep Deterministic Policy Gradient (DDPG)

DDPG is an off-policy, model-free DRL algorithm designed for continuous action spaces. It combines insights from Deep

Q-Networks (DQN) and the Actor-Critic framework. Its core components include an Actor network that maps states to deterministic actions and a Critic network that evaluates the value of these actions. To stabilize training, DDPG uses target networks for both the Actor and Critic, which are updated slowly via "soft updates." It also employs a replay buffer to store past experiences and sample them randomly for updates, breaking temporal correlations. Exploration is achieved by adding noise (e.g., Gaussian noise) to the output of the deterministic actor policy during training.

2) Proximal Policy Optimization (PPO)

PPO is an on-policy DRL algorithm renowned for its stability, ease of implementation, and sample efficiency. Unlike DDPG, PPO directly optimizes a stochastic policy and uses a novel mechanism to constrain the size of each policy update. Its key innovation is the *Clipped Surrogate Objective*, which discourages large policy changes that could lead to performance collapse. The objective function uses the ratio of probabilities between the new and old policies and clips it within a small range $[1-\epsilon, 1+\epsilon]$, ensuring that policy updates remain conservative. PPO is often used with Generalized Advantage Estimation (GAE), which provides a low-variance estimate of the advantage function. The final loss function also typically includes a value function loss and an entropy bonus to encourage exploration.

III. System Model and Problem Formulation

This section establishes the theoretical foundation of our study. We first describe a UAV-mounted RIS-assisted downlink Multi-User Multiple-Input Single-Output (MU-MISO) communication system, specifically considering the clustered nature of users in an urban environment. We then introduce a channel model that incorporates realistic physical effects, including the Phase-Dependent Amplitude (PDA) of the RIS and channel estimation errors. Finally, we formulate the joint optimization problem aimed at maximizing the system's sum-rate.

A. System Architecture

We consider a downlink MU-MISO communication system deployed in an urban environment, as depicted in Fig. 1. The system is designed to provide efficient data collection services for dispersed IoT devices. Its key components are:

- **A Base Station (BS):** Located at the center of the scene at a fixed height of 15 meters, equipped with M antennas.
- **IoT Devices:** A total of 80 single-antenna IoT devices are distributed throughout a 100×100 m^2 urban area, all situated in NLOS regions with respect to the BS.
- **User Clusters:** To simulate realistic user aggregation, the 80 IoT devices are partitioned into multiple clusters using the DBSCAN algorithm. The DRL agent's objective is to serve one user cluster at each timestep, with each service group standardized to a size of $K = 4$ users.
- **A UAV-mounted RIS:** The RIS consists of L passive reflecting elements, each with an independently controllable phase shift. The UAV maintains a fixed altitude of 15 meters and moves between a series of candidate anchor points generated by a Voronoi diagram. Our study focuses on the static scenario after the UAV has arrived at an anchor point, where the RIS is activated to serve a specific user cluster within its coverage area.

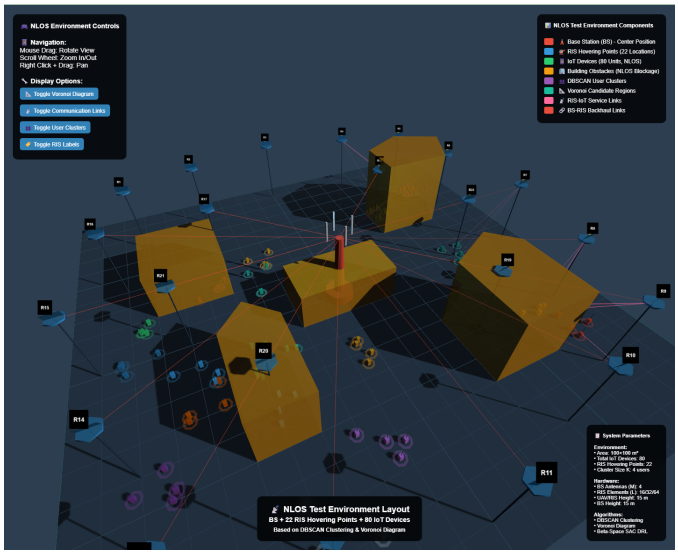


Fig. 1. System model of a UAV-mounted RIS-assisted MU-MISO communication system in an urban NLOS environment with 22 RIS anchor points and 80 IoT devices.

In this configuration, the direct link between the BS and the ground users is assumed to be completely blocked by buildings. The role of the UAV-RIS is to establish a cascaded BS-RIS-user reflection link, thereby bypassing obstacles, overcoming NLOS fading, and enhancing communication quality.

B. Channel Model

The received signal z_k at the k-th user from the BS is modeled via the cascaded BS-RIS-user channel as follows [6]:

$$z_k = \mathbf{h}_k^T \mathbf{\Phi} \mathbf{H} \mathbf{G} \mathbf{x} + w_k \tag{1}$$

where:

- $\mathbf{x} \in \mathbb{C}^{K\times 1}$ is the data stream for K users, with $\mathbb{E}[\mathbf{x}\mathbf{x}^H] = \mathbf{I}_K$.
- $\mathbf{G} \in \mathbb{C}^{M\times K}$ is the transmit beamforming (precoding) matrix at the BS.
- $\mathbf{H} \in \mathbb{C}^{L\times M}$ is the channel matrix from the BS to the RIS.
- $\mathbf{\Phi} = \text{diag}(\phi_1, \ldots, \phi_L) \in \mathbb{C}^{L\times L}$ is the diagonal reflection matrix of the RIS, where ϕ_l is the reflection coefficient of the l-th element.
- $\mathbf{h}_k \in \mathbb{C}^{L\times 1}$ is the channel from the RIS to the k-th user.
- $w_k \sim \mathcal{CN}(0, \sigma_w^2)$ is the Additive White Gaussian Noise (AWGN) at the k-th user.

The elements of channel matrices $\mathbf{H}$ and $\mathbf{h}_k$ follow a Rician fading model to capture both LoS and NLoS components. For notational simplicity, we define the cascaded channel for user k as $\mathbf{D}_k = \text{diag}(\mathbf{h}_k)\mathbf{H} \in \mathbb{C}^{L\times M}$ [6], [7]. The received signal can then be rewritten as:

$$z_k = \boldsymbol{\phi}^T \mathbf{D}_k \mathbf{G} \mathbf{x} + w_k \tag{2}$$

where $\boldsymbol{\phi} = [\phi_1, \ldots, \phi_L]^T \in \mathbb{C}^{L\times 1}$ is the RIS phase-shift vector.

1) Phase-Dependent Amplitude (PDA) Model

Ideal RIS elements only alter the phase of the signal, with a constant reflection amplitude of unity. In practice, however, the reflection amplitude co-varies with the phase shift. To model this physical limitation, we adopt the PDA model from [6]:

$$\phi_l = \beta(\varphi_l)e^{j\varphi_l} \tag{3}$$

where $\varphi_l \in [0, 2\pi)$ is the phase shift of the l-th element, and $\beta(\varphi_l)$ is the phase-dependent amplitude, given by:

$$\beta(\varphi_l) = (1 - \beta_{\min}) \left(\frac{\sin(\varphi_l - \mu) + 1}{2} \right)^{\kappa} + \beta_{\min} \tag{4}$$

Here, $\beta_{\min} \in [0, 1]$, $\mu \geq 0$, and $\kappa > 0$ are constants determined by the hardware implementation of the RIS. This model introduces additional challenges to the optimization problem, making it more aligned with real-world physics.

2) Channel Estimation Error

In a practical system, CSI is never perfectly known. We consider the presence of channel estimation errors, where the cascaded channel observed by the RL agent is $\hat{\mathbf{D}}_k$:

$$\hat{\mathbf{D}}_k = \mathbf{D}_k + \mathbf{E}_k, \quad \forall k = 1, \ldots, K \tag{5}$$

where $\mathbf{E}_k \in \mathbb{C}^{L \times M}$ is the channel estimation error matrix, whose elements are i.i.d. complex Gaussian random variables with $e_{l,m}^{(k)} \sim \mathcal{CN}(0, \sigma_e^2)$ [6].

C. Problem Formulation

The objective of this research is to jointly optimize the BS beamforming matrix $\mathbf{G}$ and the RIS phase-shift vector $\boldsymbol{\phi}$ to maximize the downlink sum-rate for a selected user cluster. This optimization is considered for a **single decision timestep** where the UAV is stationary at an anchor point, serving a specific user group $\mathcal{K} = \{1, 2, \ldots, K\}$.

Given $\mathbf{G}$ and $\boldsymbol{\phi}$, the Signal-to-Interference-plus-Noise Ratio (SINR) for the k-th user within cluster $\mathcal{K}$ is:

$$\text{SINR}_k = \frac{|\boldsymbol{\phi}^T \mathbf{D}_k \mathbf{g}_k|^2}{\sum_{j \in \mathcal{K}, j \neq k} |\boldsymbol{\phi}^T \mathbf{D}_k \mathbf{g}_j|^2 + \sigma_w^2} \tag{6}$$

where $\mathbf{g}_k$ is the k-th column vector of matrix $\mathbf{G}$. The achievable rate for user k is $R_k = \log_2(1 + \text{SINR}_k)$. The system sum-rate R_Σ for the served cluster $\mathcal{K}$ is the sum of all user rates within that cluster:

$$R_\Sigma(\mathcal{K}) = \sum_{k \in \mathcal{K}} R_k = \sum_{k=1}^{K} \log_2 \left(1 + \frac{|\boldsymbol{\phi}^T \mathbf{D}_k \mathbf{g}_k|^2}{\sum_{j \in \mathcal{K}, j \neq k} |\boldsymbol{\phi}^T \mathbf{D}_k \mathbf{g}_j|^2 + \sigma_w^2} \right) \tag{7}$$

Here, both k and j are indices of users within the currently served cluster $\mathcal{K}$, where $|\mathcal{K}| = K = 4$. The outer summation aggregates the rates of all users in the group, while the inner summation accounts for the intra-group interference.

Our optimization problem (P1) is formulated as:

$$\begin{aligned} \max_{\mathbf{G}, \boldsymbol{\phi}} \quad & R_\Sigma(\mathcal{K}) \\ \text{s.t.} \quad & C1 : \phi_l = \beta(\varphi_l)e^{j\varphi_l}, \quad \forall l = 1, \ldots, L \\ & C2 : \text{tr}(\mathbf{G}\mathbf{G}^H) \leq P_t \end{aligned} \tag{8}$$

where C1 is the non-ideal PDA model constraint for the RIS, and C2 is the maximum transmit power constraint at the BS. This is a non-convex optimization problem with highly coupled variables, for which we employ DRL to find an effective solution.

IV. Methodology and Training Framework

To address the high-dimensional, non-convex joint optimization problem defined in Chapter III, we designed a comprehensive intelligent network deployment and control framework based on DRL. This section details the four core components of this framework: (1) a smart city environment generation mechanism based on DBSCAN and Voronoi diagrams; (2) the design of the DRL agent's state, action, and reward spaces; (3) the various DRL algorithms evaluated; and (4) an innovative virtual user complementation mechanism to stabilize training.

A. Smart City Environment Generation Mechanism

Traditional DRL applications in communications often rely on purely random channel matrix generation for training, which can lead to poor generalization of the learned policies. To overcome this limitation, we developed an environment generation mechanism that more closely reflects realistic deployment logic.

1) **Scenario Layout and NLOS Device Generation:** In a 100×100 m^2 virtual city, we first generate a fixed set of random polygonal and rectangular buildings. Subsequently, 80 IoT devices are deployed, strictly constrained to be within the NLOS region of the BS to simulate the most challenging signal blockage scenarios.
2) **DBSCAN User Clustering:** To model the natural clustering of users, we employ the DBSCAN algorithm to group the 80 devices. Based on a neighborhood radius ($\epsilon = 8$m) and a minimum sample count (min_samples=1), geographically proximate devices are assigned to the same cluster. To match the input dimensions of our DRL model, any cluster with more than four devices is further subdivided, ensuring each final user group contains at most four devices.
3) **Voronoi Diagram for Candidate Point Generation:** To identify optimal potential locations for RIS deployment, we utilize Voronoi diagrams. The diagram is constructed using the centroids of buildings and the BS location as seeds. The edges of the Voronoi cells, representing paths equidistant from multiple obstacles, and their vertices naturally serve as excellent candidate locations for deploying the aerial platform.
4) **Candidate Point Filtering and Service Assignment:** We extract a large set of candidate points from Voronoi vertices and LoS/NLoS boundary points. These points are rigorously filtered to ensure that final anchor points are within the LoS of the BS, are not inside buildings, and are within a 50-meter radius of the BS. Finally, the system assigns an optimal RIS anchor point to each user cluster to provide the best possible coverage.

B. Reinforcement Learning Framework Design

We model the optimization problem as a Markov Decision Process (MDP), with the following elements:

- **State Space $\mathcal{S}$:** The state vector s_t is a concatenation of the previous action a_{t-1}, the transmit power allocated to and received power at each user, and the real and imaginary parts of the estimated cascaded channel matrix $\hat{\mathbf{D}}_k$. The total state dimension is $2(MK + L) + 2K + 2KLM$. To

enhance learning efficiency, the state vector at each timestep undergoes whitening.

- **Action Space $\mathcal{A}$:** The action a_t consists of the real and imaginary parts of the beamforming matrix $\mathbf{G}$ and the RIS phase-shift vector $\boldsymbol{\phi}$. The total action dimension is $2(MK+L)$. The raw output from the neural network is normalized to satisfy the total power constraint at the BS and the unit modulus constraint for each RIS element's phase shift.

  Physical Meaning of the Action Vector: An End-to-End Joint Control Mechanism

  It is crucial to emphasize that the beamforming matrix $\mathbf{G}$ in this framework is not predetermined by traditional signal processing algorithms (e.g., Zero-Forcing). Instead, determining the specific values of both $\mathbf{G}$ and $\boldsymbol{\phi}$ is the core task learned by the DRL agent. The agent's Actor network acts as a unified decision-making center, operating as follows:

  1) **Perception and Decision:** At each step, the agent observes the current state, which includes the channel information, and feeds it into the Actor network. The network, through its complex non-linear transformations, directly outputs a single, long "action" vector of dimension $2(MK+L)$.
  2) **Action Decoding and Execution:** The environment receives this action vector and decodes it. The first $2MK$ elements are reshaped into the complex beamforming matrix $\mathbf{G}$ for the BS, while the subsequent $2L$ elements are used to configure the RIS phase-shift vector $\boldsymbol{\phi}$.
  3) **Learning and Emergent Behavior:** The agent does not possess any innate understanding of antenna theory or electromagnetic physics. It learns entirely by maximizing the reward function (i.e., the sum-rate). Through tens of thousands of trial-and-error iterations, the agent gradually discovers that for a given channel state, outputting specific values for $\mathbf{G}$ (which result in constructive interference toward the RIS) and specific values for $\boldsymbol{\phi}$ (which precisely reflect the energy to the users) yields the highest reward.

  Thus, the BS's "energy focusing" and the RIS's "signal reflection" are not two separate steps but a single, highly coordinated decision learned by the DRL agent within a unified action space. This end-to-end learning approach enables the system to discover superior joint control strategies that are often intractable for conventional, decoupled optimization methods.
- **Reward Function R:** The reward function directly reflects our optimization objective. At each timestep t, after the agent executes action a_t, the environment calculates the corresponding sum-rate R_Σ using (7), which serves as the instantaneous reward r_t.

C. Deep Reinforcement Learning Algorithms

This study conducts a comprehensive performance evaluation of several mainstream and advanced DRL algorithms.

- **Soft Actor-Critic (SAC):** An off-policy algorithm based on the maximum entropy reinforcement learning framework. We evaluate two variants:
  - **SAC (Golden):** The agent has access to perfect CSI and is unaffected by the PDA effect, serving as a theoretical upper bound on performance.
  - **SAC (Mismatch):** The agent only observes CSI with errors, and the environment includes the PDA effect.
- **Beta-Space Exploration SAC:** To mitigate the performance degradation in SAC (Mismatch) due to model discrepancies, we employ this enhanced algorithm. It introduces an additional "Explorer" network that is trained to maximize prediction error, guiding the agent to explore regions of the state-action space that are poorly understood due to the model mismatch, thereby implicitly learning a more accurate model of the true environment.
- **Deep Deterministic Policy Gradient (DDPG):** A classic off-policy algorithm that learns a deterministic policy and uses additive noise for exploration. It serves as an important off-policy baseline for comparison with SAC.
- **Proximal Policy Optimization (PPO):** A leading on-policy algorithm known for its training stability and robustness. It constrains policy updates using a clipped surrogate objective. As a representative on-policy method, PPO provides a valuable contrast to the off-policy algorithms.

D. Virtual User Complementation Mechanism

In our environment generation process, the DBSCAN clustering can result in user groups smaller than our standard size ($K=4$). Directly feeding variable-dimension states into a DRL model would lead to training instability or failure. To resolve this, we designed an innovative **Virtual User Complementation Mechanism**.

This mechanism works as follows: when an RIS anchor point serves a group with $K_{real} < K$ real users, the system automatically creates $K-K_{real}$ "virtual users." The geographical locations of these virtual users are set to be identical to the location of the real user in that group who is closest to the RIS. From the DRL agent's perspective, it is always serving a group of $K=4$ users, ensuring that the dimensions of its state input and action output vectors remain constant. This mechanism elegantly transforms a problem with a dynamically changing number of users into a fixed-dimension learning problem, significantly enhancing the training stability and applicability of the DRL model across different user group sizes. This is a key methodological innovation of our research.

V. Experimental Results and Analysis

This section presents and analyzes the results of our simulation experiments. We evaluate the convergence performance of the different DRL models during training and conduct an in-depth investigation of their performance in an NLOS-dominant urban environment across various RIS sizes and anchor points, thereby validating the effectiveness of our proposed framework.

A. Experimental Parameters

Our experimental parameters are based on the system model and standard configurations for the DRL algorithms, as detailed in Table I and Table II.

TABLE I
SIMULATION ENVIRONMENT AND SYSTEM PARAMETERS

Parameter	Value
Environment Parameters	
City Size	$100 \times 100\ m^2$
Total IoT Devices	80
BS Position	(50, 50, 15) m
RIS/UAV Altitude	15 m
DBSCAN Radius (ϵ)	8 m
DBSCAN Min Samples	1
Communication Parameters	
BS Antennas (M)	4
Users per Group (K)	4
RIS Elements (L)	16, 32, 64
BS Transmit Power (P_t)	30 dBm (1 W)
AWGN Variance (σ_w^2)	1×10^{-2}
Channel Estimation Error Var. (σ_e^2)	1×10^{-2}
PDA Model Parameters	
Min. Amplitude (β_{min})	0.3 (L=16), 0.6 (L=32, 64)
Amplitude Attenuation Factor (κ)	1.5

TABLE II
DRL ALGORITHM HYPERPARAMETERS

Common Hyperparameters	Value
Learning Rate (Actor & Critic)	1×10^{-3}
Network Hidden Layers	2 layers, 256 neurons each (ReLU)
Total Training Timesteps	20,000
Off-Policy (SAC, DDPG) Parameters	
Replay Buffer Size	20,000
Batch Size	16
Target Network Update Rate (τ)	1×10^{-3}
SAC Entropy Coefficient (α)	0.2 (Auto-tuned)
On-Policy (PPO) Parameters	
Steps per Update (n_steps)	2048
Update Epochs (n_epochs)	10
Clipping Range (clip_range)	0.2
Entropy Coefficient (ent_coef)	0.01

B. Analysis of Training Convergence

Figures 2, 3, and 4 show the moving average reward (sum-rate) convergence curves for the five DRL models during training with RIS sizes of L=16, 32, and 64, respectively.

From these curves, we observe several key trends:

- **Clear Performance Hierarchy:** Across all RIS scales, the final performance of the algorithms shows a distinct hierarchy. The Beta-Space SAC and the two standard SAC variants significantly outperform DDPG and PPO. This indicates that for the high-dimensional RIS control problem with non-ideal effects, the maximum entropy-based SAC framework is more advantageous than traditional DDPG and PPO.
- **Superiority of Beta-Space Exploration:** Beta-Space Exploration SAC consistently achieves the best performance among algorithms that account for hardware mismatch. For instance, with $L = 32$ (Fig. 3), its final converged performance reaches $9.414\ bps/Hz$, substantially outperforming not only SAC (Mismatch) at $7.982\ bps/Hz$ but also the ideal SAC (Golden) at $7.301\ bps/Hz$. This strongly suggests that its Explorer network successfully adapts to the PDA effect and discovers control strategies superior even to those found under an ideal model.
- **Performance of DDPG and PPO:** While DDPG and

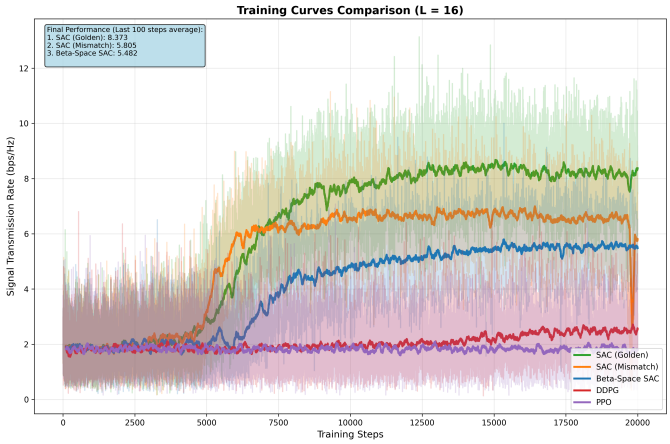


Fig. 2. Training convergence curves for DRL models ($L = 16$).

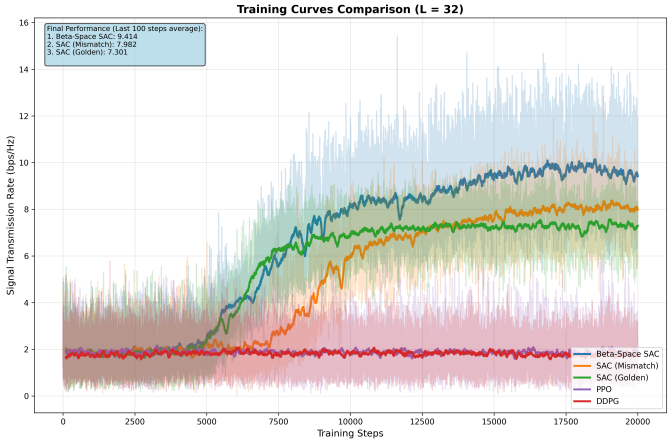


Fig. 3. Training convergence curves for DRL models ($L = 32$).

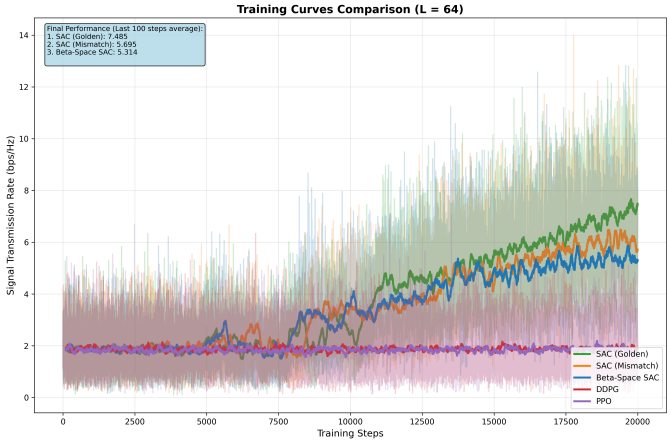


Fig. 4. Training convergence curves for DRL models ($L = 64$).

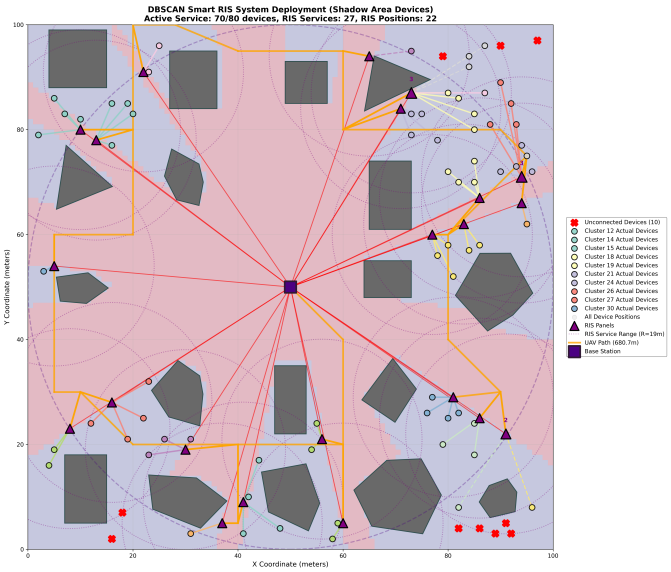


Fig. 5. Layout of the NLOS test environment, showing the BS, 22 RIS anchor points, and 80 clustered IoT devices.

PPO do converge, their final performance is considerably lower than the SAC family, typically stabilizing around $2-3$ bps/Hz. This may be attributed to the inefficiency of DDPG's exploration mechanism (deterministic policy plus noise) and PPO's on-policy sampling in navigating the complex RIS phase space.

C. Performance Analysis in NLOS Scenario

To evaluate the performance of each algorithm in a realistic deployment scenario, we conducted tests in a fixed NLOS environment composed of 22 RIS anchor points. In this setup, there is no direct LoS between the users and the BS; communication relies entirely on the RIS reflections. Figure 5 illustrates the layout of this test environment.

1) Performance Comparison at Individual Anchor Points

Figures 6, 7, and 8 present the average sum-rate achieved by each algorithm at 10 representative anchor points for L=16, 32, and 64, respectively.

These charts illustrate the performance disparities at specific geographic locations:

- **Consistent Advantage of SAC Family:** At nearly all anchor points, the performance of the three SAC variants (light blue, cyan, teal) far exceeds that of DDPG (purple), PPO (orange), and the random phase baseline (pink). This demonstrates the general applicability of the SAC framework to the RIS phase optimization problem.
- **Exceptional Performance of Beta-Space SAC:** Beta-Space SAC (light blue) achieves the highest performance in the vast majority of cases. For example, with $L = 64$ (Fig. 8) at anchor point RIS-7, it achieves a sum-rate of 4.4 bps/Hz, significantly higher than SAC (Mismatch) at 4.2 bps/Hz and DDPG at 1.6 bps/Hz. This again confirms its robustness in handling hardware impairments.
- **Impact of RIS Scale:** Comparing the three figures, it is evident that as the number of RIS elements increases from 16 to 64, the performance of all algorithms improves significantly, which is consistent with the physical principle of higher beamforming gains from larger antenna arrays.

2) Average Performance Analysis Across All Anchor Points

To draw statistically significant conclusions, we calculated the average performance across all 22 anchor points. The results are plotted in Figures 9, 10, and 11.

These average performance charts provide the definitive conclusions of our study:

- **Overwhelming Advantage of Beta-Space SAC:** Across all RIS scales, Beta-Space SAC demonstrates the highest average performance. In the most representative case of $L = 64$ (Fig. 11), it achieves an average sum-rate of 3.35 ± 0.51 bps/Hz.
- **Quantitative Comparison with Baselines** ($L = 64$)**:**
  - Compared to **SAC (Mismatch)** (2.68 ± 0.50 bps/Hz), Beta-Space SAC shows a performance improvement of approximately **25.0%**.
  - Compared to **DDPG** (1.16 ± 0.61 bps/Hz), the performance gain is approximately **188.8%**.
  - Compared to **PPO** (1.13 ± 0.51 bps/Hz), the performance gain is approximately **196.5%**.
  - Compared to the **Random Phase** baseline (0.91 ± 0.24 bps/Hz), the performance gain is approximately **268.1%**.
- **Robustness of a Conclusion:** This performance ranking holds true for $L = 16$ and $L = 32$ as well, indicating that the superiority of Beta-Space SAC is not coincidental but a robust characteristic across different RIS scales. The standard deviation (error bars) also reflects the stability of each algorithm across different geographical locations, with the SAC family generally exhibiting a smaller relative standard deviation than DDPG and PPO, suggesting better generalization capabilities.

D. Summary of Results

The experimental results in this chapter systematically validate our hypotheses across three levels of analysis: training convergence, location-specific performance, and overall average performance. In the realistic, NLOS-dominant urban environment designed with DBSCAN user clustering, the Beta-Space Exploration SAC algorithm, specifically engineered to handle hardware impairments, demonstrates comprehensive superiority. It not only exhibits high learning efficiency but also consistently delivers the optimal sum-rate across all test scenarios, significantly outperforming all baseline methods, including standard SAC, DDPG, and PPO.

VI. Conclusion and Future Work

A. Conclusion

This study has conducted an in-depth investigation into the communication challenges of UAV-mounted RIS systems for IoT data collection in smart cities. By integrating an advanced user clustering mechanism with a comparative evaluation of multiple DRL algorithms, we have successfully established an intelligent and environmentally adaptive phase control framework. The main conclusions are summarized as follows:

1) **A structured UAV-RIS simulation framework was successfully constructed and validated:** Our novel introduc-

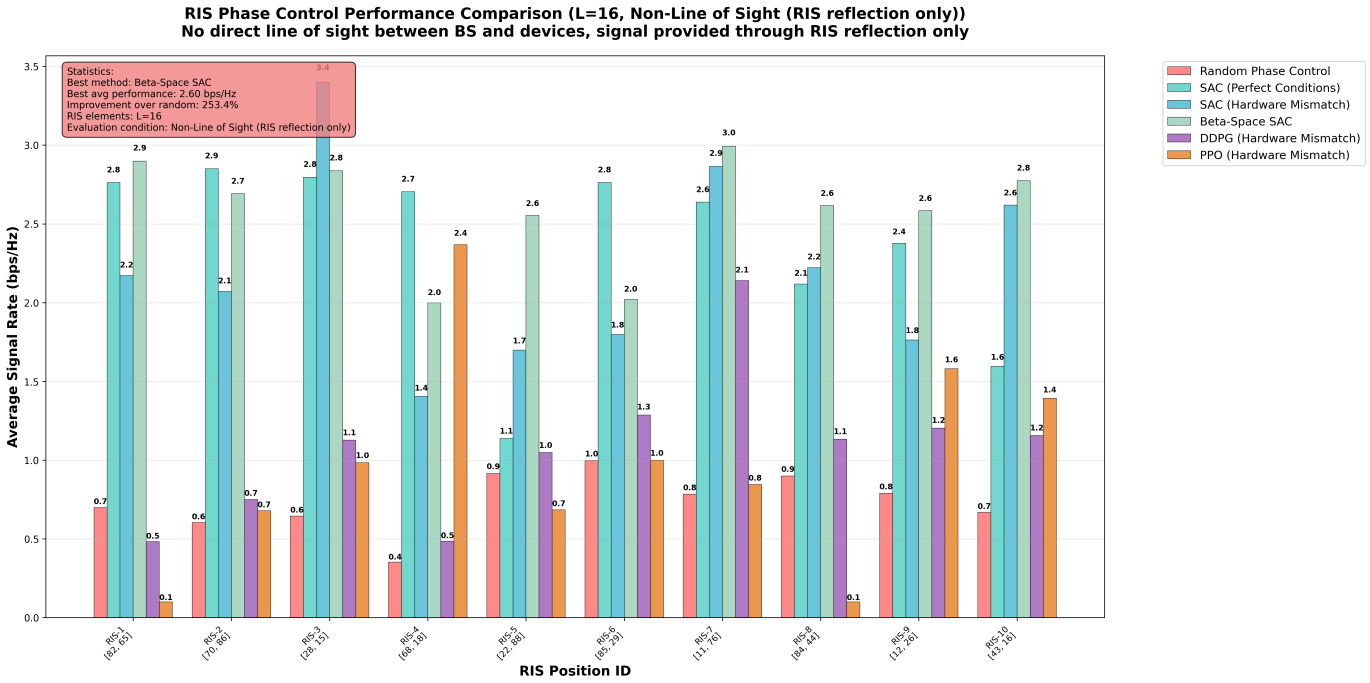


Fig. 6. Comparison of sum-rate performance at individual anchor points for $L = 16$ in the NLOS scenario.

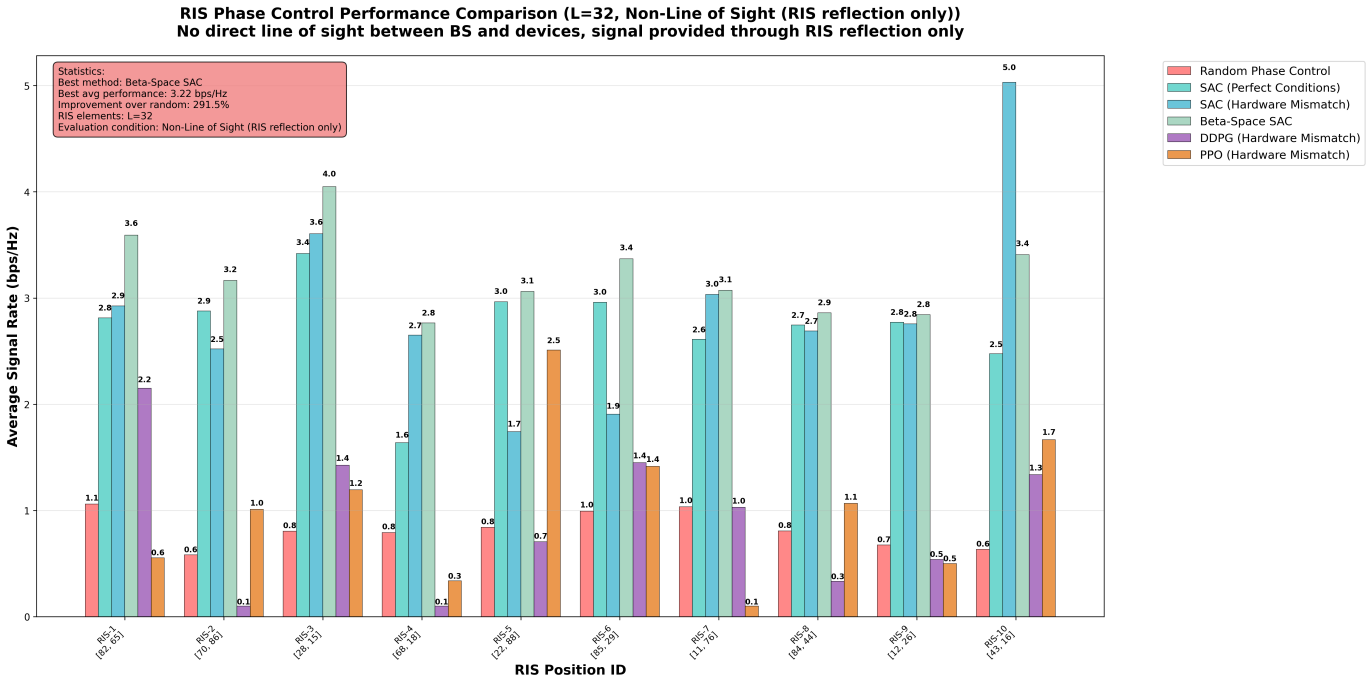


Fig. 7. Comparison of sum-rate performance at individual anchor points for $L = 32$ in the NLOS scenario.

tion of DBSCAN-based user clustering and Voronoi-based anchor point generation makes the simulation environment more faithful to the clustered distribution of users in real cities. This provided the DRL agent with more meaningful, structured learning tasks, effectively enhancing the specificity of the training and the practical utility of the final policy.

2) **A clear performance hierarchy among DRL algorithms was established:** In our challenging NLOS-dominant scenarios, experimental results unequivocally show that the maximum entropy-based SAC family of algorithms significantly outperforms traditional DDPG and PPO for solving the RIS phase optimization problem.
3) **The Beta-Space Exploration SAC algorithm demonstrated exceptional performance and robustness:** When accounting for realistic hardware non-idealities like PDA

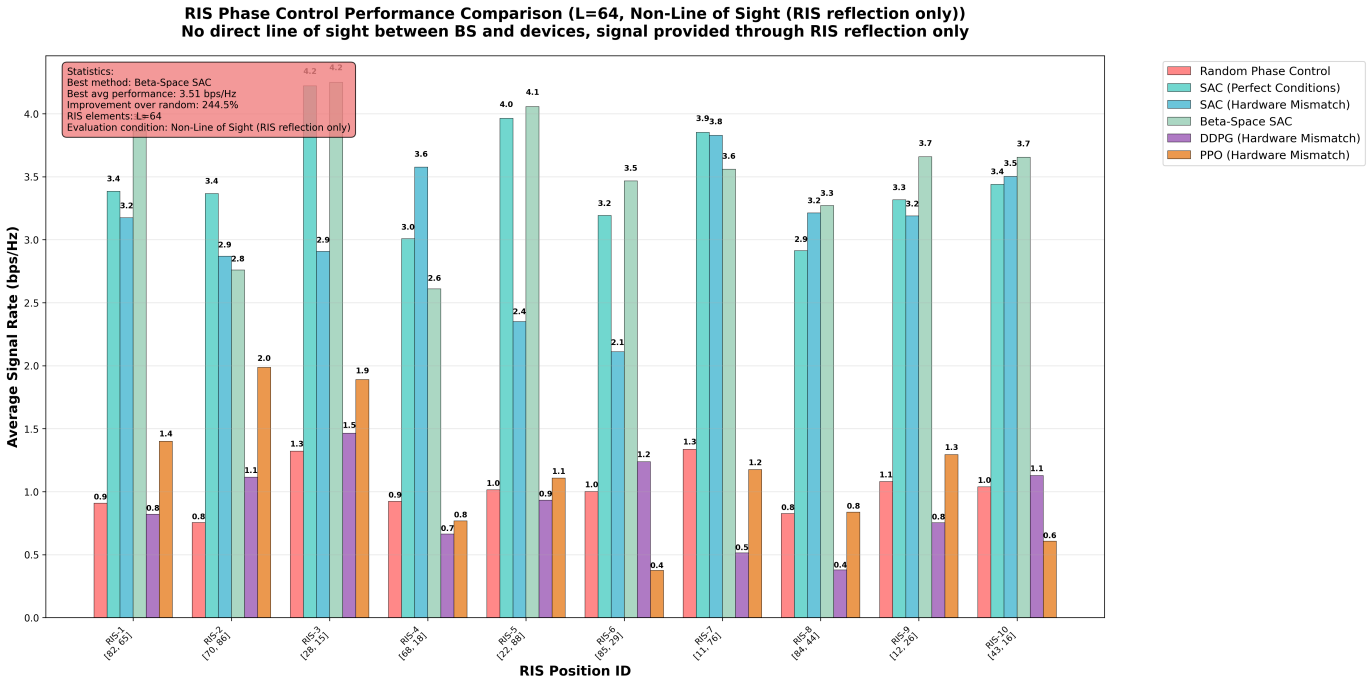


Fig. 8. Comparison of sum-rate performance at individual anchor points for $L = 64$ in the NLOS scenario.

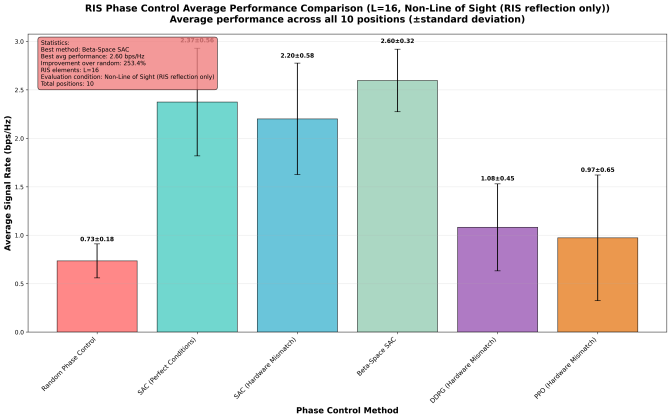


Fig. 9. Average performance comparison across all 22 anchor points ($L = 16$, NLOS). Error bars represent standard deviation.

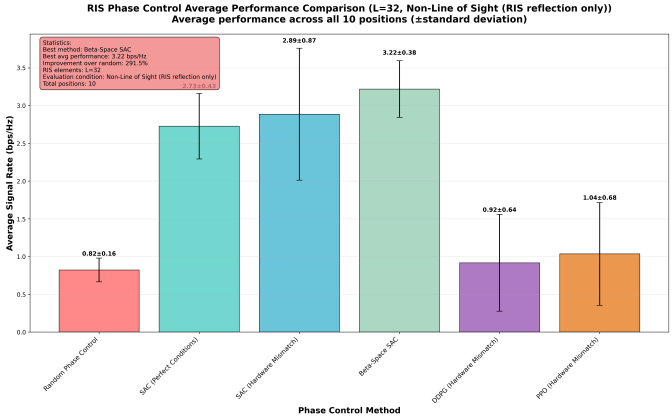


Fig. 10. Average performance comparison across all 22 anchor points ($L = 32$, NLOS). Error bars represent standard deviation.

and imperfect CSI, the Beta-Space Exploration SAC algorithm showed an overwhelming advantage. Its average sum-rate far surpassed all benchmarks, proving that its Explorer network can effectively learn and compensate for real-world hardware limitations, making it a powerful solution for model mismatch problems.

4) **The "Virtual User Complementation Mechanism" proved effective:** This mechanism successfully resolved the issue of inconsistent DRL model input dimensions caused by dynamic user group sizes. It ensured stable model training and represents a key technical innovation for applying DRL to such practical deployment scenarios.

In summary, this research not only validates the application potential of UAV-RIS systems in smart city data collection but also provides a complete and pragmatic solution pipeline—from environment generation and algorithm comparison to performance evaluation. It offers a solid technical foundation and empirical data for the adaptive deployment of future reconfigurable intelligent communication networks.

B. Future Work

Building upon the findings of this study and the latest academic frontiers, we propose several structured and forward-looking directions for future research.

- **Integrated Sensing and Communication (ISAC)-RIS Systems:** The next logical step is to expand the UAV's role from a pure communication platform to a dual-function platform with sensing capabilities. Future work could redesign the DRL framework to jointly optimize the communication sum-rate and the sensing Cramér-Rao Lower Bound (CRLB) for specific targets, exploring the trade-offs and synergies between these two functions [30].

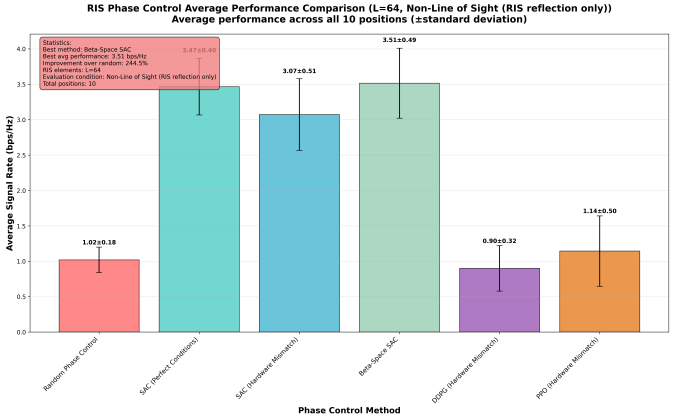


Fig. 11. Average performance comparison across all 22 anchor points ($L = 64$, NLOS). Error bars represent standard deviation.

- **Task-Oriented Semantic-Aware RIS Systems:** A fundamental shift would be to move the optimization objective from the physical layer's "throughput" to the application layer's "task effectiveness." A key research question is how to design RIS phase control policies that maximize the semantic decoding accuracy or Quality of Experience (QoE) at the receiver, rather than just the SINR [31]. This could involve end-to-end joint training of the RIS controller and a semantic decoder.
- **Resilient and Proactive Anti-Jamming RIS Systems:** Future critical applications necessitate resilience against adversarial actions. A challenging research direction is to train a DRL agent to control the RIS to perform active anti-jamming by forming nulls in the direction of a jammer while enhancing the signal for legitimate users. This could be modeled as a dynamic game between the UAV-RIS agent and a jamming agent, solved using game-theoretic MADRL frameworks [32].

In conclusion, the integration of UAVs and RIS opens up vast possibilities for future wireless communications. By incorporating sensing capabilities, semantic efficiency, and adversarial resilience, we can move towards building truly intelligent, autonomous, and robust next-generation air-ground integrated networks.

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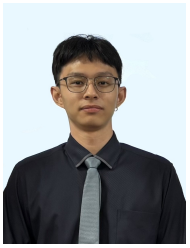
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